

Support Vector Machine (SVM) is a powerful **supervised learning algorithm** used for **classification and regression**. It works by finding the optimal hyperplane that maximizes the margin between two classes.

1. Intuition Behind SVM

SVM aims to find a decision boundary that **best separates** two classes while maximizing the **margin** (distance between the closest points of each class, called support vectors).

Hyperplane & Margin

- The **hyperplane** is the decision boundary separating classes.
- The **margin** is the perpendicular distance from the closest points (support vectors) to the hyperplane.
- **Goal**: Maximize the margin to improve generalization.

Mathematical Formulation

For a dataset with **features** X and labels y , the hyperplane equation is:

$$w^T x + b = 0$$

where:

- w = weight vector (normal to the hyperplane)
- b = bias term
- x = input feature vector

For correct classification:

$$y_i(w^T x_i + b) \geq 1$$

where $y_i \in \{+1, -1\}$.

2. Types of SVM

(a) Linear SVM

- Used when data is **linearly separable**.

- Finds the **optimal hyperplane** that maximizes the margin.

Linear SVM Optimization (Hard Margin)

$$\min_{w,b} \frac{1}{2} \|w\|^2$$

subject to:

$$y_i(w^T x_i + b) \geq 1, \quad \forall i$$

Soft Margin SVM

If data is **not perfectly separable**, we introduce **slack variables** ξ to allow some misclassifications:

$$y_i(w^T x_i + b) \geq 1 - \xi_i, \quad \forall i$$

where C is the regularization parameter controlling trade-off between margin size and misclassification.

(b) Non-Linear SVM (Kernel Trick)

When data is **not linearly separable**, SVM maps it to a **higher-dimensional space** using the **kernel trick**.

Popular Kernels

1. Polynomial Kernel:

$$K(x, x') = (x^T x' + c)^d$$

2. Radial Basis Function (RBF) Kernel (Most Used):

$$K(x, x') = \exp(-\gamma \|x - x'\|^2)$$

- γ (**gamma**): Controls the influence of each training example.
- **Higher γ** $\rightarrow$ more complex boundary (risk of overfitting).

3. Sigmoid Kernel (Similar to Neural Networks):

$$K(x, x') = \tanh(\alpha x^T x' + c)$$

3. SVM for Regression (SVR)

SVM can also be used for **regression** by allowing a tolerance ϵ around the true value.

$$|y_i - (w^T x_i + b)| \leq \epsilon$$

Popular **SVR** Kernels:

- **Linear SVR**: Good for simple linear relationships.
- **RBF SVR**: Works well for complex, non-linear regression tasks.